

Summary and conclusion

Data models in database systems

Lecture 10

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RTU

**DATORZINĀTNES UN
INFORMĀCIJAS
TEHNOLOĢIJAS FAKULTĀTE**

Reminder of the course grading

- **50%: grade for the practical work**
 - minimum grade (4) if the assignment scores 50% and meets the minimum requirements
 - must receive **at least 12 grade points** for all the practical works (each max. 10 grade points) to get access to exam
- **25%: the midterm test**
 - minimum passing grade (4) on ~~50%~~ 40% of the points.
 - midterm must be passed
- **25%: the final exam**
 - minimum passing grade (4) at 50% of the points.
 - final exam must be passed

Course policies

- No do-overs, except the last exam (as per RTU policy) and once – midterm test
- If
 - the grade points for the practical works are not enough to pass the course,
 - **or** midterm test is failed

student has failed the course before the exam

Application for repeated midterm test



URL

Link to register for midterm test retake 

Please note that you will not be allowed to retake the midterm test if you are not registered by **January 5, 2026.**



1. I consent that by filling this form I annul my previous grade.

Regardless of my score in the repeated midterm test, it will become my score for midterm test. If you select "I disagree", you will not be registered for the midterm test. *

☐ I agree

☐ I disagree

The plan for the remaining semester

Week 16	Final lecture	PW3 submission- deadline on 28.12.
Week 17	Repeated midterm test – 08.01.	Registration for repeated midterm test – deadline on 05.01. By January 11 at latest you will know if you are taking final exam
Week 18	Final exam – 16.01. 10:00, Z8-302	15.01. 14:30 – remote consultation
Week 19	Repeated final exam – 22.01., 14:30	January 20, 14:30-16:00, Z10-404 you will be able to see your exam paper

Final exam – general information

- January 16, 10:00, Zundas 8 – 302
- The exam will be 90 minutes
- **You need to meet semester requirements to take the exam**
- The difficulty and structure are similar to the midterm test
- The final exam is summative
- More information about the structure on the remote consultation on January 15, 14:30

The Development of Database Data Models



Directions of DB development

- Managing complex data
- Managing very big data
- Increasing the value of data



**There is a need for a complex
solution to employ data at it's
best value**

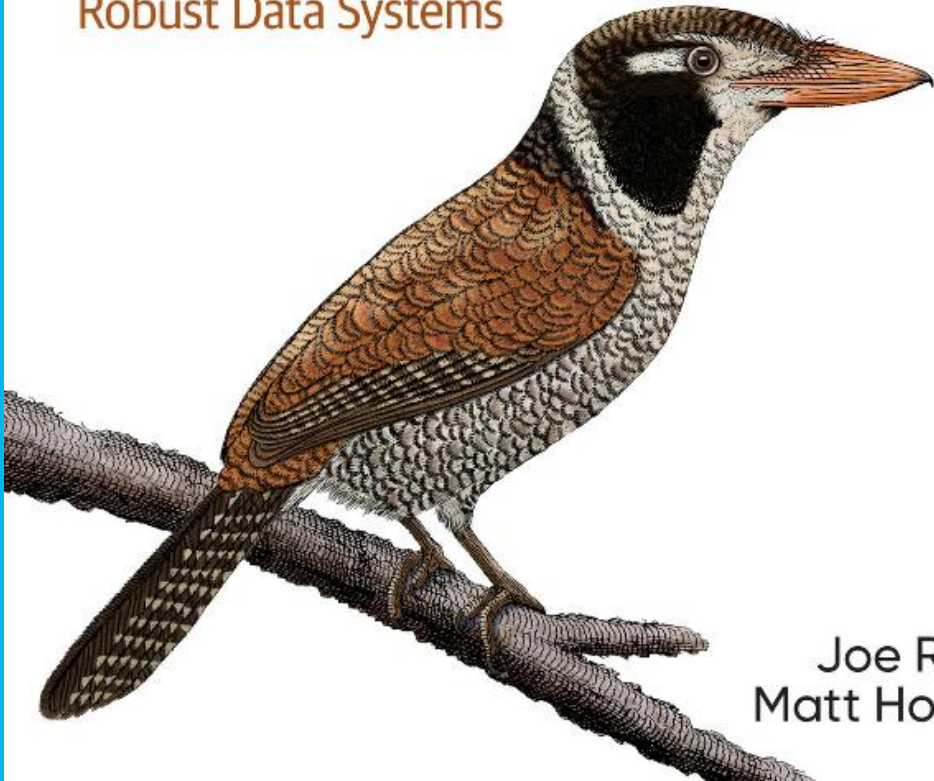
Data engineering



O'REILLY®

Fundamentals of Data Engineering

Plan and Build
Robust Data Systems

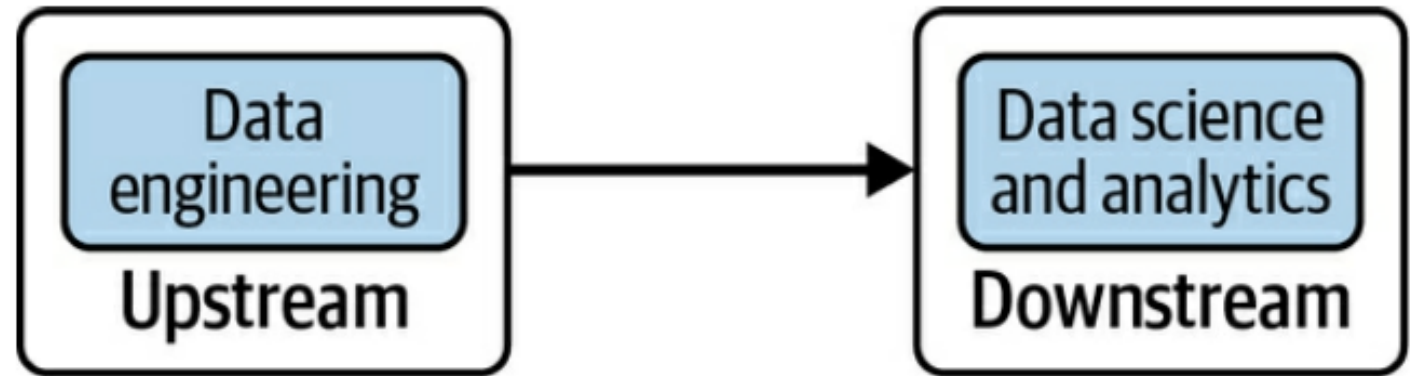


Joe Reis &
Matt Housley

Reis, Joe; Housley, Matt. Fundamentals
of Data Engineering: Plan and Build
Robust Data Systems. 2022. O'Reilly
Media,

Data management related areas

- Data analytics
- Data engineering
- Data science
- ...and many more



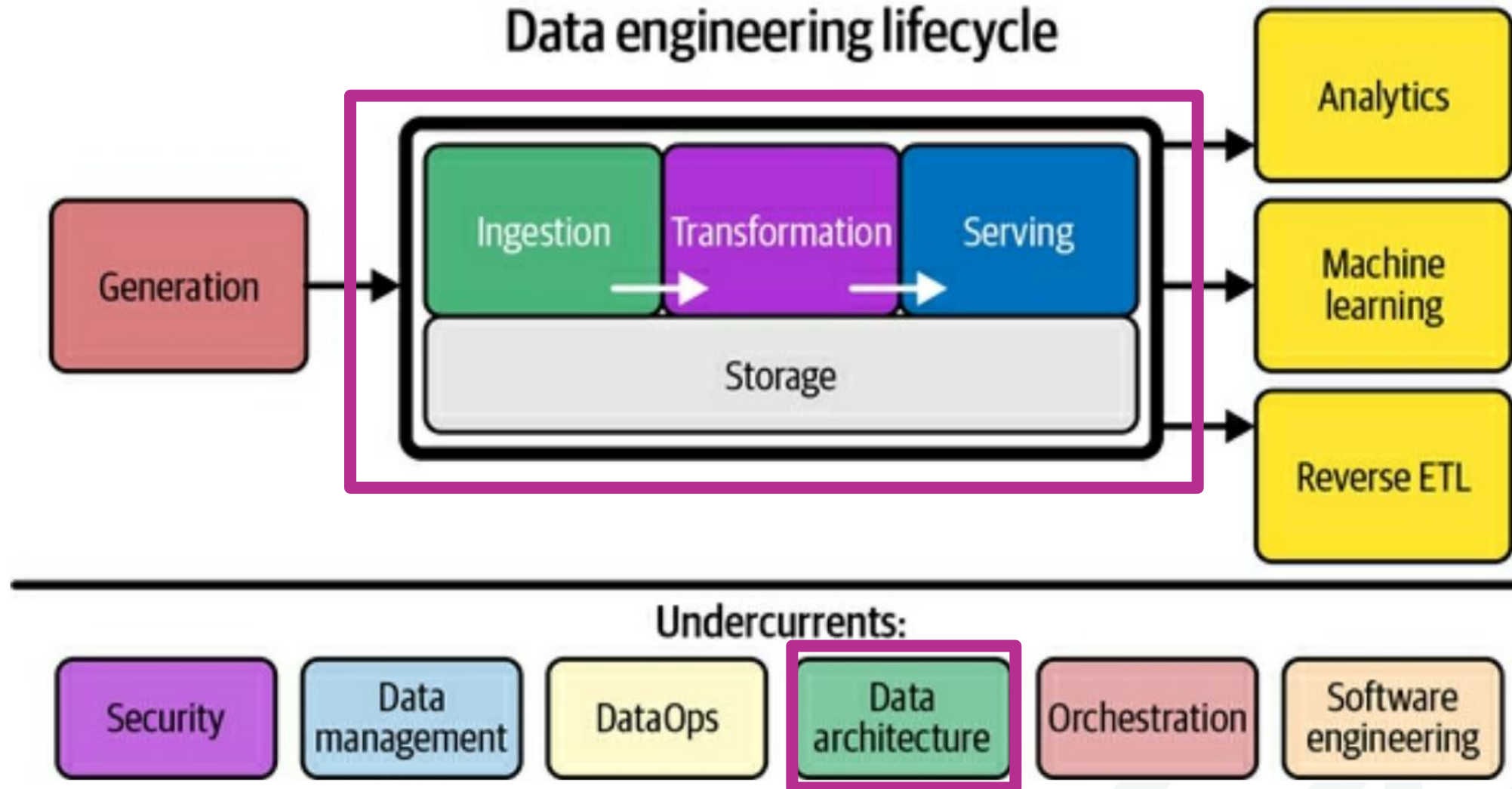
What is data engineering?

- «Data engineering is the **development, implementation, and maintenance of systems** and processes that **take in raw data and produce high-quality, consistent information** that supports downstream use cases, such as analysis and machine learning.»

Data engineering in course context

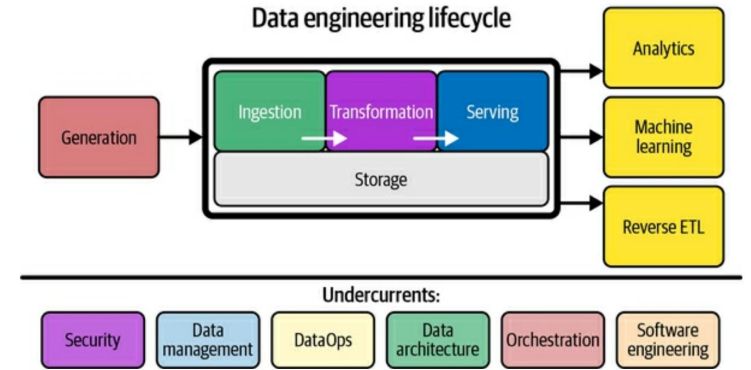
- This is a step up from relational databases
- RDBS can support a single task, within data engineering platform we are talking about robust, potentially multi-model solution for analytics.
- In this course we have touched some of the parts of data engineering, namely, some **data architectures**, as well as **multiple data models**

Data engineering lifecycle



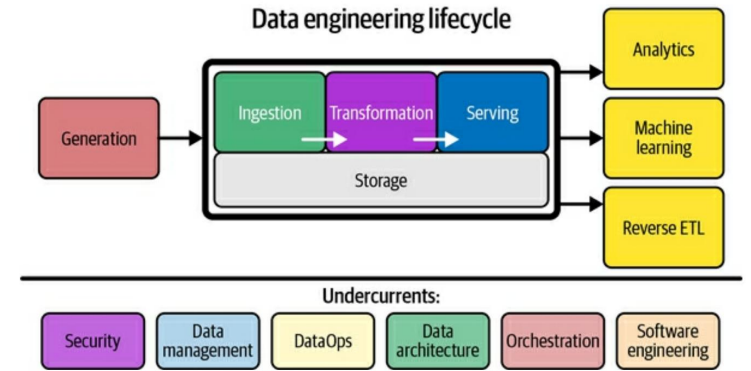
Generation and source systems

- Source system is the origin of data
 - an IoT device, an application message queue, or a transactional database
- No control over source system in a data engineering process



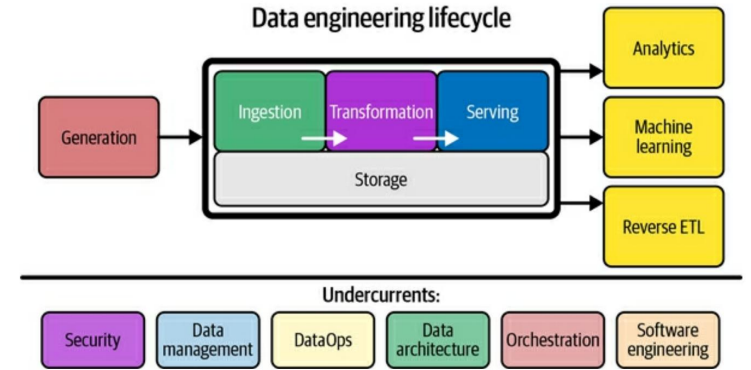
Storage systems

- Usually there is more than one storage solution
 - Cloud data warehouse + streaming framework
- Selection depends on your use cases, data volumes, frequency of ingestion, format, and size of the data being ingested
- No one-size-fits-all approach



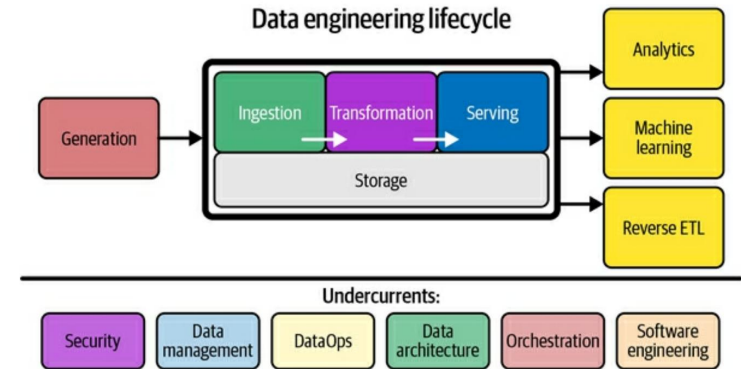
Ingestion process

- Similar to extraction in data warehousing (that is one of the options)
- In general, source systems produce data continuously
- One can ingest data as a **stream** or in **batches**

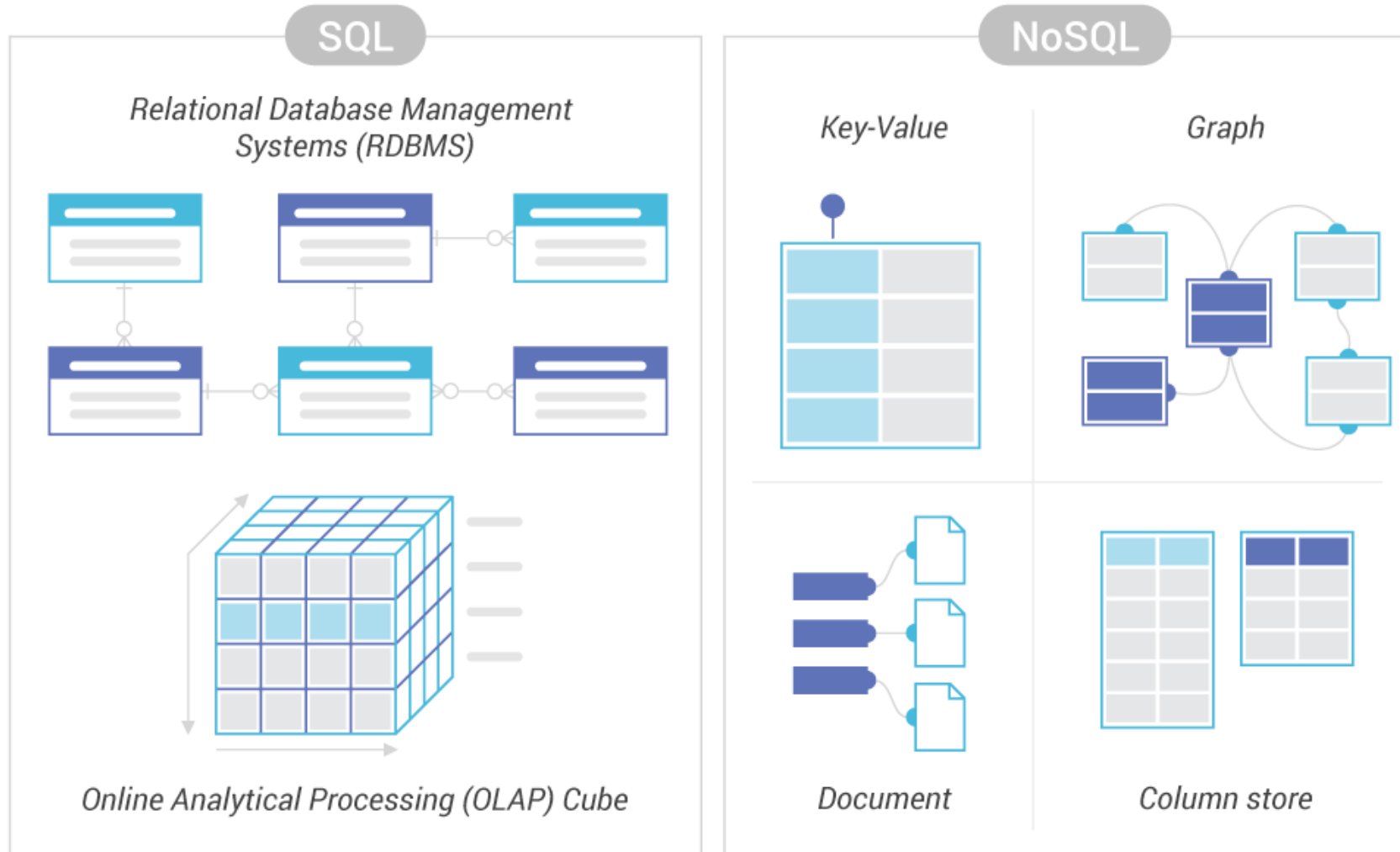


Transformation

- This is where data model is created
- A **data model** represents the way data relates to the real world
- Data must be structured and standardized to
 - Match business and represent workflow
 - For better processing
- There are many options



Different models

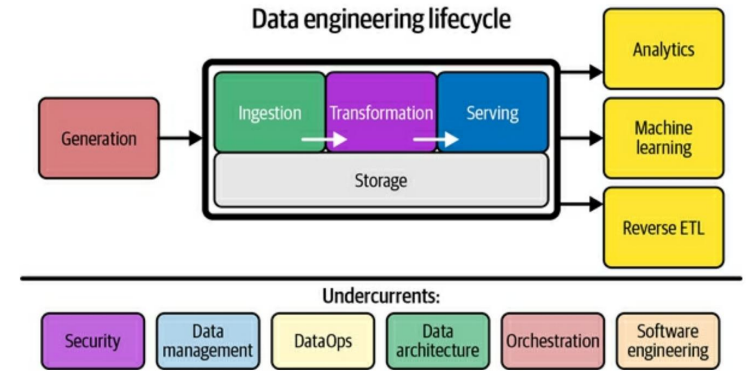


When to choose different models?

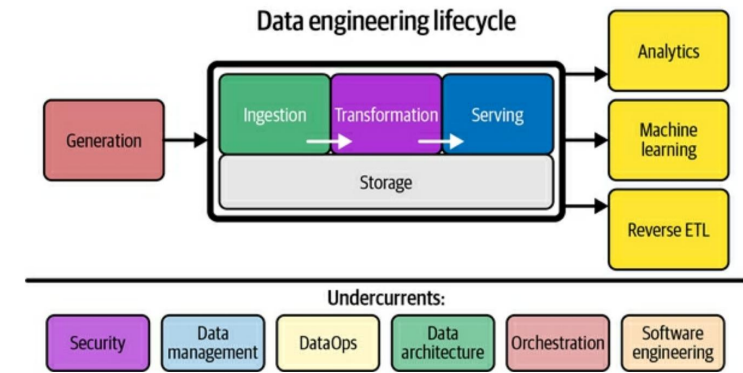
- Column store – used primarily for faster analytics
- ORDB – used to ensure better connectivity and improve processing speed in edge cases
- Semi-structured store (hierarchical data model) – used for stream processing (schema-on-read), for storing data with non-specific schema
- Note that it all is related, including the architecture and storage structure

Serving and analytics

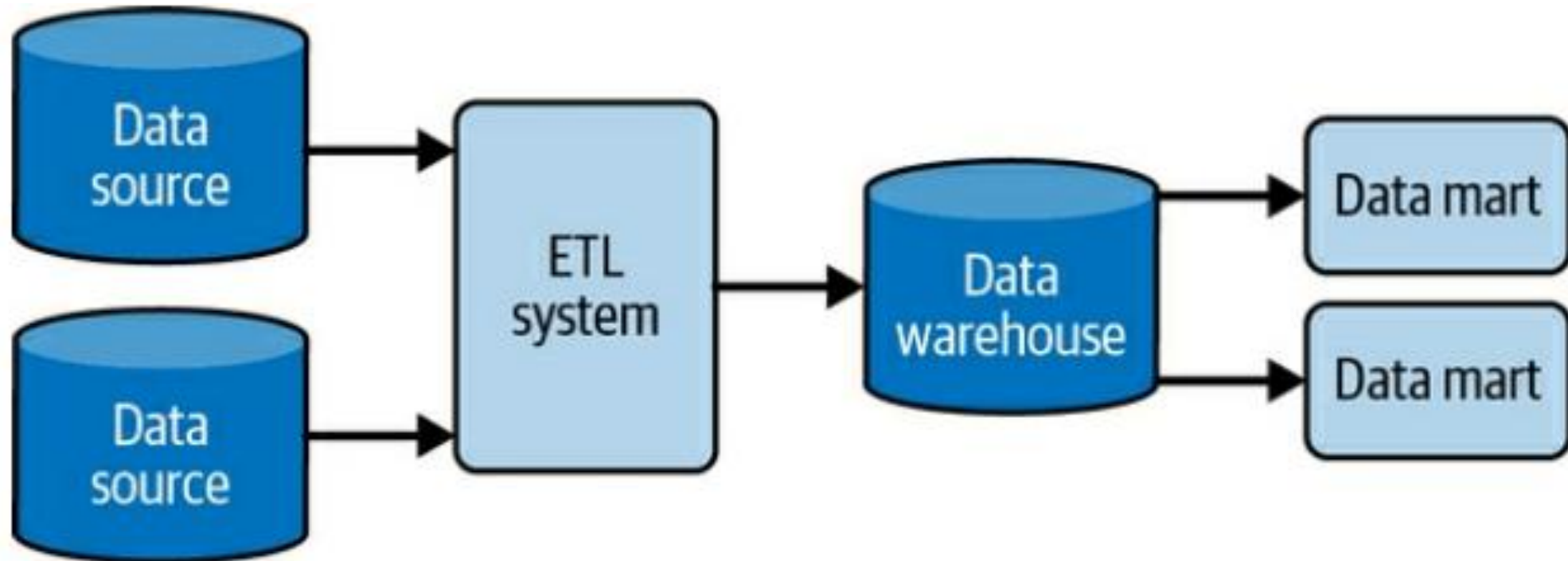
- Some of the popular uses of data
 - analytics
 - ML
 - reverse ETL
- ML engineering and data engineering boundaries can be fuzzy
 - Who extract metadata and featurizes?
 - AI-embedded databases
- Especially with new AI-enabled stores
 - Feature stores (e.g. vector databases)



Types of architectures: data warehouse



- Data warehouse is one of the oldest and most well-established data architectures



When is this architecture suitable

- As I mentioned, it all is related – you cannot really «separate» these concepts
- Data warehousing is architecture that can employ multiple models, typically
 - Relational model
 - Column store
- Data warehouse is a powerful architecture for aggregations
- Not suitable if you have
 - Service-oriented approach as opposed to monolith architecture
 - Incoming data stream
 - Large amounts of unstructured data
- There are other options if those properties apply, e.g., lambda/kappa architecture, data mesh, data lake/lakehouse etc.

Future trends

- Less complexity, more easy-to-use data tools, stress on tool integration
- Cloud-scale data
- More business-involved data engineering – data mesh is an example of this trend
- Blurring database and application – i.e., more in-database features

Thank you!

